

MSC MECHANICAL ENGINEERING

Semester Project

Numerical modelling of multiphase flow processes in
porous media

Student:

Villa César Jacques Hanakea Fomalhaut Melchior
cvilla@ethz.ch - 22-801-708

Baltaian Nicholas
nbaltaian@ethz.ch - 22-818-058

Project Supervised by:

Prof. Angst Ueli and Dr. Luka Malenica

Institute for Building Materials

June 12, 2026

Contents

1	Introduction	4
1.1	Motivation	4
1.2	Project Objectives	4
2	Theoretical Considerations and Methodology	5
2.1	Flow Physics and Definitions	5
2.1.1	Surface Tension, Contact angle and Wettability	5
2.1.2	Young-Laplace equation and capillary pressure	6
2.1.3	Governing Equations	6
2.2	Computational Methodology	7
2.2.1	Volume Of Fluid	7
2.2.2	Finite Volume Method	7
2.2.3	Courant Number	8
2.2.4	Software	9
3	Results and Data Analysis	9
3.1	Verification Cases on simple Geometries	9
3.1.1	Capillary Rise in a Tube	9
3.1.2	Wettability Effects on Trapped Bubble	13
3.2	Complex Porous Media Geometries	15
3.2.1	Bentheimer Sandstone	15
3.2.2	Effects of Surface Wettability	16
3.2.3	Water VS IPA	20
4	Discussion	24
4.1	Interpretation	24
4.1.1	Impact of Surface Wettability	24
4.1.2	Role of Fluid Properties (Water vs. IPA)	24
4.2	Assumption and Source of Errors	24
5	Conclusions	25
	References	27
6	Appendix	28
6.1	Project Contributions	28
6.2	Supplementary Simulation Interface Frames	29
6.3	Capillary Rise	29
6.4	Effect of Wettability	32

List of Figures

1	Illustration of molecular force asymmetry at a fluid-fluid interface. Source: [Booyabazooka, 2008]	5
2	Illustration of different contact angles and their corresponding wetting regimes. Adapted from the CSC Scientific Blog [Gatenby, 2016].	6
3	Schematic setup of the cylindrical capillary rise validation benchmark. h denotes the rise height, θ the contact angle, p the gauge pressure and p_{liq} the wetting fluid's pressure	10
4	Temporal evolution of a capillary rise interface profile ($\theta = 20^\circ$, $r = 1$ mm)	11
5	Temporal evolution of a trapped bubble against a solid wall ($\theta = 140^\circ$), plotted alongside the corresponding capillary pressure jump at the interface.	13
6	Extracted $50 \times 140 \times 125$ voxel subdomain used for analyzing microscopic wetting dynamics and contact angle variations along the 140×125 walls.	15
7	Extracted $80 \times 140 \times 80$ voxel subdomain used for investigating the impact of liquid property variations between two distinct fluids.	16
8	Temporal evolution of water rise along the front steel wall for a contact angle ($\theta = 60^\circ$)	17
9	Temporal evolution of water rise along the front steel wall for a contact angle ($\theta = 120^\circ$)	18
10	Temporal evolution of water saturation (S_w) at the steel boundary walls for hydrophilic ($\theta = 60^\circ$) and hydrophobic ($\theta = 120^\circ$) contact angles.	19
11	Initial conditions for the simulations	20
12	Temporal evolution of Water within Bentheimer Sandstone	21
13	Temporal evolution of IPA within Bentheimer Sandstone	22
14	Comparison of the transient liquid saturation S_{liquid} as a function of simulation time for the water (blue) and IPA (green)	23
15	Supplementary simulation interface configurations ($\theta = 40^\circ$, $r = 1$ mm), highlighting the smaller inertial overshoot due to lower surface tension force	29
16	Supplementary simulation interface configurations ($\theta = 60^\circ$, $r = 1$ mm)	30
17	Temporal evolution of a capillary rise interface profile ($\theta = 20^\circ$, $r = 0.5$ mm)	31
18	Temporal evolution of water rise along the back steel wall for a contact angle ($\theta = 60^\circ$)	34
19	Temporal evolution of water rise along the back steel wall for a contact angle ($\theta = 120^\circ$)	35

Abstract

This semester project investigates multiphase flow processes within porous media. Using GeoChem-Foam, an open-source computational fluid dynamics (CFD) framework for pore-scale simulations, we simulated fluid infiltration through a complex micro-CT scan of Bentheimer Sandstone. Our simulations were driven by two primary objectives:

- Observe wetting dynamics and gas-liquid distribution on steel boundary walls at the microscopic level.
- Assess how the distinct properties of water and isopropyl alcohol impact fundamental fluid behaviors, such as capillary dynamics, within complex porous media.

To evaluate wetting dynamics, two distinct cases of boundary conditions were simulated by modifying the contact angle of the steel walls: a hydrophilic case at 60° and a hydrophobic case at 120° . The findings reveal that despite the microscopic scale and the governing influence of capillary pressure, wettability severely affects the overall distribution of the fluid. Saturation measurements, which quantify the percentage of the total pore area filled with water, showed a significantly lower water saturation of 42% for the hydrophobic surface (120°), compared to 75% for the hydrophilic surface (60°). These results demonstrate that increasing the contact angle directly translates to pronounced hydrophobic behavior along the solid surface, ultimately restricting the liquid infiltration capacity of the porous medium.

To evaluate how intrinsic fluid properties dictate transport mechanisms within a porous medium, we compared the invasion dynamics of water and isopropyl alcohol (IPA). Saturation measurements revealed significantly greater water saturation (85%) in the domain compared to IPA (76%). This distinct saturation dynamic highlights the profound impact of surface tension and kinematic viscosity on flow behavior. Water, possessing a higher surface tension and lower kinematic viscosity, generates a larger capillary pressure gradient (P_c) and experiences reduced flow resistance. This results in a rapid, quasi-linear initial invasion, stabilizing in just 0.01 s. Conversely, IPA exhibits a slower accumulation, taking nearly three times longer to stabilize (0.035 s) due to increased viscous drag and weaker capillary forces. Furthermore, the 9% deficit in the final saturation plateau for IPA suggests that this less favorable balance of forces leads to minor localized trapping or bypassing of defensive phase pockets within smaller pore throats.

1 Introduction

1.1 Motivation

In cement-based materials, moisture transport significantly affects the durability of reinforced concrete structures due to humidity, which allows for the corrosion of steel reinforcements in concrete. Despite its importance, moisture transport in porous materials remains an ongoing challenge.

Traditionally, fluid infiltration in porous media is modeled using capillary tube dynamics. Washburn (1921) derived the basic relationship for capillary flow, which is driven by surface tension and opposed by viscous drag [Washburn, 1921]. These baseline models were later refined to include dynamic contact angles, inertial effects, and transient regimes [Zhud et al., 2000].

However, predicting moisture transport in real cement structures is more complex than in idealized tubes. Experiments show that moisture imbibition in cement matrices exhibits anomalous transport behaviors. Conventional capillary models cannot fully capture these effects. To address this, recent models use kinetic transport coefficients, such as time/saturation-dependent permeability, to simulate anomalous moisture uptake [Zhang and Angst, 2020].

To model local transport phenomena, modern research uses 3D microstructural characterization. Imaging techniques, such as X-ray Micro-Computed Tomography (micro-CT), are used to quantify the pore size distribution, connectivity, and complex geometry of the materials [K et al., 2021].

The aim of this project is to numerically investigate multiphase flow processes at the pore scale within complex porous media. This research is broadly motivated by the durability challenges inherent to reinforced porous media, where moisture transport and the subsequent degradation of embedded reinforcement play a critical role. To better understand these underlying mechanisms, this project focuses on fundamental pore-scale flow behaviors. By comparing the capillary-driven infiltration of aqueous and non-aqueous liquids into sandstone, this study evaluates how distinct fluid properties dictate transport processes and final pore-space occupancy.

1.2 Project Objectives

The primary objective of this project is to simulate a pore-scale multiphase Computational Fluid Dynamics (CFD) model using the GeoChemFoam solver platform [Maes and Menke, 2026], which is compiled as an extension of the open-source OpenFOAM CFD architecture [OpenCFD Ltd., 2022], in order to isolate and quantify how variations in surface wettability and variations in fluid properties govern gas-liquid distribution.

To achieve this, the investigation is divided into two distinct objectives:

- Analyze microscopic wetting dynamics and contact angle variations along a boundary wall. The objective is to compare the behavior of a fluid along a steel wall with two separate contact angles between the steel plate and water.
- Investigate the impact of fluid property variations on multiphase flow. The objective is to systematically modify the physical properties of the invading liquid phase within GeoChemFoam, comparing two distinct air-fluid combinations to determine their direct effect on pore-scale transport behavior.

This report follows a logical progression from fundamental theory to applied results. Chapter 2 establishes the theoretical foundation of multiphase flows, identifying the core challenges inherent in CFD simulations for these phenomena. Building on this context, Chapter 3 details the computational methods employed to verify the solver's accuracy and presents the results obtained from simulating flow through complex porous geometries. These findings are further evaluated in Chapter 4, paving the way for Chapter 5, which concludes the study.

2 Theoretical Considerations and Methodology

2.1 Flow Physics and Definitions

Multiphase flow refers to the simultaneous flow of different phases (solid, liquid, or gas) or distinct chemical properties that remain separated by a physical boundary, known as an interface. In the context of this project, we focus on immiscible fluids, meaning the two fluids do not mix and maintain a distinct separation.

2.1.1 Surface Tension, Contact angle and Wettability

Surface Tension, denoted as σ [N/m], is the elastic-like property of a liquid's surface that allows it to resist external forces. It arises from an imbalance of the forces of attraction (or cohesive forces) acting on the molecules at the interface between phases. Away from the surface, a molecule is surrounded by neighboring molecules, each applying an equal cohesive force to one another; hence, this results in a net-zero force balance. At the interface, however, molecules are no longer in a homogeneous environment; instead, they are exposed to a mixture of molecules from both phases. This causes an imbalance in the cohesive forces acting on the interfacial molecules and puts the surface molecules into a state of tensile stress. This tensile stress is the physical explanation of surface tension.

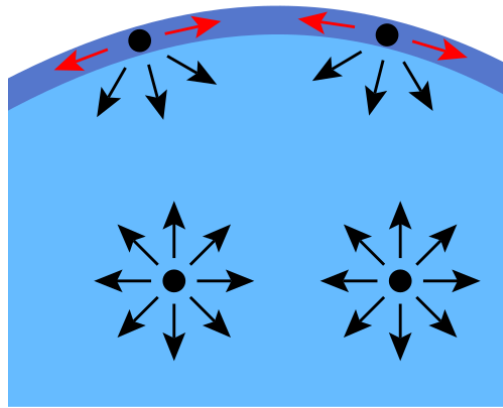


Figure 1: Illustration of molecular force asymmetry at a fluid-fluid interface. Source: [Booyabazooka, 2008]

The contact angle, denoted as θ [°], is defined as the angle formed at the intersection where two immiscible fluids meet a solid surface.

This angle acts as a direct indicator of the surface's wettability, which dictates how a liquid behaves when in contact with a solid surface. Depending on the measured contact angle, the wettability behavior can be classified into several regimes. It is important to note that these angular thresholds are approximations and serve as general guidelines:

- **Wetting:** Typically defined for angles less than 75° (or broadly below 90°). In this regime, the liquid has a high affinity for the solid and tends to spread over the surface. When we are in the presence of water, we talk about a hydrophilic surface.
- **Neutral wettability:** Occurs when the contact angle is approximately between 75° and 105°. The surface does not show a dominant preference for either fluid.

- **Non-wetting:** Characterized by angles greater than 105° (or broadly above 90°). Here, the liquid minimizes its contact with the surface, forming distinct droplets. For water, we talk about hydrophobic surface.

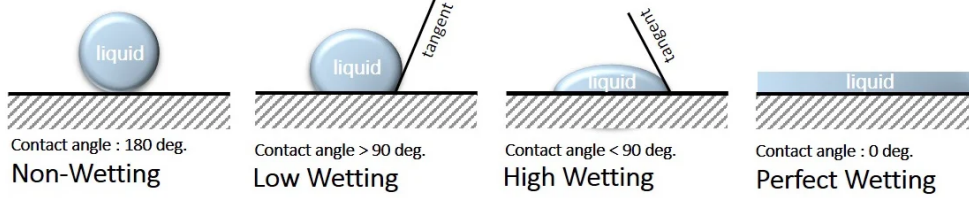


Figure 2: Illustration of different contact angles and their corresponding wetting regimes. Adapted from the CSC Scientific Blog [Gatenby, 2016].

2.1.2 Young-Laplace equation and capillary pressure

As stated in subsection 2.1.1, at the interface, a surface tension force acts as a barrier, pulling the molecules together. When a fluid interface is perfectly planar, the cohesive forces act strictly parallel to the surface, maintaining a horizontal equilibrium where the net perpendicular force is zero. However, when the interface is forced to curve, this symmetry is broken. Due to the curvature, the molecules at the surface will be pulled inward, towards the center of curvature. This additional normal force acting on the curved surface will push on the fluid and create an internal pressure. This curvature-induced pressure difference results in a localized pressure discontinuity across the phase boundary, a phenomenon macroscopically defined as capillary pressure (P_c). The equation that quantifies the pressure jump is called the Young-Laplace equation. In its most generalized form, the relation balances the pressure jump against the interfacial tension and the geometric curvature of the meniscus:

$$P_c = P_{nw} - P_w = \Delta p = \sigma \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad (2.1)$$

where σ represents the surface tension, P_{nw} is the pressure of the non-wetting fluid, P_w is the pressure of the wetting liquid, and R_1 and R_2 are the principal radii of curvature.

2.1.3 Governing Equations

To accurately model and track the motion of these two fluids, we rely on the fundamental governing equations of fluid dynamics. Specifically, we solve the Navier-Stokes equations, which consist of the mass conservation (continuity) equation and the momentum equation, incorporating an additional term to represent the interfacial surface tension force. Assuming incompressible flow, the continuity equation and the momentum equation are expressed as:

$$\nabla \cdot \vec{u} = 0 \quad (2.2)$$

$$\frac{\partial(\rho \vec{u})}{\partial t} + \nabla \cdot (\rho \vec{u} \vec{u}) = -\nabla p + \nabla \cdot \left[\mu (\nabla \vec{u} + (\nabla \vec{u})^T) \right] + \rho \vec{g} + \vec{f}_{\text{surface tension}} \quad (2.3)$$

Where ρ is the density, $\vec{u}$ is the velocity vector, p is the pressure, μ is the dynamic viscosity, and $\vec{g}$ represents gravitational acceleration. The final term, $\vec{f}_{\text{surface tension}}$, is relevant for capillary-driven flows as it models the forces acting at the fluid-fluid interface.

2.2 Computational Methodology

2.2.1 Volume Of Fluid

While the Navier-Stokes equations provide a continuous physical description, they must be discretized to be solved numerically. We discretize the continuous domain into a finite number of small control volumes, or meshes. To track the interface between the two immiscible fluids within this discretized domain, we employ the Volume of Fluid (VOF) method.

Various methods exist for interface capturing, such as geometric or algebraic VOF, Level-Set, front-tracking or hybrid CLSVOF [Moukalled et al., 2016]. GeoChemFoam utilizes an algebraic VOF approach. The core principle of VOF is to assign a new scalar variable, typically denoted as the phase fraction α , to each cell in the grid. Specifically, $\alpha = 1$ indicates that the cell is completely filled with the primary fluid (e.g., water), $\alpha = 0$ indicates the cell is filled with the secondary fluid (e.g., air), and values of $0 < \alpha < 1$ signify that the cell contains the interface between the two fluids. To dynamically track the interface, an additional transport equation for the scalar α must be solved alongside the governing fluid equations:

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \vec{u}) = 0 \quad (2.4)$$

Once this α field is obtained, the effective physical fields for density ρ and viscosity μ can be defined across the computational domain using an algebraic volumetric average:

$$\rho = \alpha \rho_1 + (1 - \alpha) \rho_2 \quad \text{and} \quad \mu = \alpha \mu_1 + (1 - \alpha) \mu_2 \quad (2.5)$$

Furthermore, this phase fraction field is used to determine the shape and curvature of the interface. The normal vector of the interface $\vec{n}$ and the mean curvature κ are calculated based on the gradient of the phase fraction.

$$\kappa = -\nabla \cdot \vec{n} \quad \text{with} \quad \vec{n} = \frac{\nabla \alpha}{|\nabla \alpha|} \quad (2.6)$$

A significant advantage of this algebraic VOF approach is that the mass conservation can be satisfied if it is discretized correctly.

2.2.2 Finite Volume Method

The finite volume method is a numerical discretization technique used to solve partial differential equations. In the finite volume method, volume integrals in a partial differential equation that contain a divergence term are converted to surface integrals, using the divergence theorem. These surface integrals are then evaluated as fluxes across the boundaries of each individual control volume, or cell, within the computational mesh. Because the flux leaving a given cell through a shared face is identical to the flux entering the adjacent cell, the finite volume method is inherently conservative. The basic idea behind this framework is to use Gauss's theorem to simplify how we track the physical properties within each cell. Instead of tracking whatever changes are happening within the volume of each cell, this approach allows us to only follow the sum of what crosses the cell's outer boundaries. Mathematically, Gauss's theorem, states that the volume integral of the divergence of a vector field $\vec{\Phi}$ is identically equal to the net surface integral of that field passing through the bounding faces:

$$\int_V \nabla \cdot \vec{\Phi} \, dV = \oint_S \vec{\Phi} \cdot \vec{n} \, dA \approx \sum_f \vec{\Phi}_f \vec{S}_f \quad (2.7)$$

where $\vec{n}$ is the unit normal vector pointing outward from the cell surface area A , and $\vec{S}_f$ is the discrete area vector for each individual cell face f .

To apply this numerical framework to multiphase flow, the governing partial differential equation (2.3) is integrated over a discrete control volume V_P :

$$\begin{aligned} \int_{V_P} \frac{\partial(\rho \vec{u})}{\partial t} dV_P + \int_{V_P} \nabla \cdot (\rho \vec{u} \vec{u}) dV_P = \int_{V_P} -\nabla p dV_P + \int_{V_P} \nabla \cdot [\mu(\nabla \vec{u} + (\nabla \vec{u})^T)] dV_P \\ + \int_{V_P} \rho \vec{g} dV_P + \int_{V_P} \sigma \kappa \nabla \alpha dV_P \end{aligned} \quad (2.8)$$

By applying Gauss's divergence theorem to the spatial derivatives, the continuous differential vectors governing fluid motion, pressure gradients, and viscous shear are converted directly into discrete algebraic balances across the bounding faces (f) of the cell. Using the approximation highlighted in equation (2.7), we finally obtain:

$$\begin{aligned} \frac{(\rho \vec{u})_P^{n+1} - (\rho \vec{u})_P^n}{\Delta t} V_P + \sum_f (\rho_f \vec{u}_f \cdot \vec{S}_f) \vec{u}_f = - \sum_f p_f \vec{S}_f + \sum_f [\mu_f (\nabla \vec{u} + (\nabla \vec{u})^T)_f] \cdot \vec{S}_f \\ + \rho_P \vec{g} V_P + \sum_f \sigma \kappa_f \alpha_f \vec{S}_f \end{aligned} \quad (2.9)$$

Where the subscript P denotes values evaluated at the center of the control volume cell and the subscript f represents values interpolated at the cell faces.

By repeating this discretization process for every single cell across the computational grid, the local algebraic balances are assembled into a global system of linear equations over the entire domain:

$$[\mathbf{A}][\mathbf{U}] = [\mathbf{B}] \quad (2.10)$$

where $[\mathbf{A}]$ represents the global coefficient matrix containing the convective, viscous, and mesh-dependent discretization terms, $[\mathbf{U}]$ is the vector of unknown velocity fields to be solved for at the new time step ($n+1$), and $[\mathbf{B}]$ is the source vector containing the explicit contributions from gravity, history terms, and the localized Young-Laplace capillary forces. This system of equations is solved iteratively at each time step to compute the velocity field, which is subsequently used to calculate the pressure field via the continuity equation (Equation 2.2).

For a more extensive overview of the finite volume framework, the reader is referred to *The finite volume method in Computational Fluid Dynamics: an advanced introduction with OpenFOAM and Matlab* by [Moukalled et al., 2016].

2.2.3 Courant Number

When we look at the numerical stability of a convection-diffusion equation, the chosen time step Δt is fundamentally limited by specific dimensionless parameters, as dictated by von Neumann stability analysis. The most critical parameter for convective multiphase flows is the Courant-Friedrichs-Lewy (CFL) number, often referred to simply as the Courant number.

The Courant number physically represents the ratio of the distance traveled by the fluid during one single time step to the characteristic spatial size of the mesh cell. Mathematically, it is defined as:

$$Co = \frac{|u \Delta t|}{\Delta x} \quad (2.11)$$

where u is the local fluid velocity, Δt is the time increment, and Δx represents the mesh dimension. To achieve proper mesh configuration and numerical stability, particularly when employing an explicit time discretization scheme, the time step must be strictly controlled. The underlying principle is that the fluid (and consequently the tracked phase interface) should not cross more than one adjacent

cell within a single time increment. Therefore, for explicit schemes to remain numerically stable and accurately capture the flow dynamics, the Courant number must be kept below a critical threshold which is $Co < 0.5$.

2.2.4 Software

For this project, the numerical simulations were performed using GeoChemFoam (GCF). GCF is an OpenFOAM-based framework specifically developed for pore-scale multiphase flow simulations in image-based porous media, such as micro-CT-derived geometries.

While standard OpenFOAM solvers, such as `interFoam`, are designed for general multiphase applications and often suffer from numerical instabilities at the micro-scale, GCF provides an optimized solution. In particular, it implements a more stable curvature calculation. This improvement significantly reduces spurious currents and ensures a more accurate representation of the capillary forces governing fluid transport at the pore scale.

3 Results and Data Analysis

3.1 Verification Cases on simple Geometries

In order to ensure that our simulations yield accurate results, we need to subject them to a verification step. Verification is the process of comparing the numerical results obtained through simulation to actual analytical results representing the real-world physics.

As is common in porous media characterization, exact analytical solutions are not available for complex, disordered microstructures. Consequently, the physical accuracy of the finite volume formulation must be benchmarked using simplified, idealized geometries where closed-form solutions exist. For multiphase capillary transport, we accomplished our validation through two separate tests. Namely:

- Capillary rise in a tube
- Wettability effects on trapped bubble

3.1.1 Capillary Rise in a Tube

The simulated system consists of a 2D vertical cylinder of diameter $2r$, initially filled with a non-wetting fluid (air) and placed in contact with a reservoir of wetting fluid (water) at its base. The overall geometry, coordinate system, and boundary conditions for this benchmark validation case are illustrated in the schematic layout below:

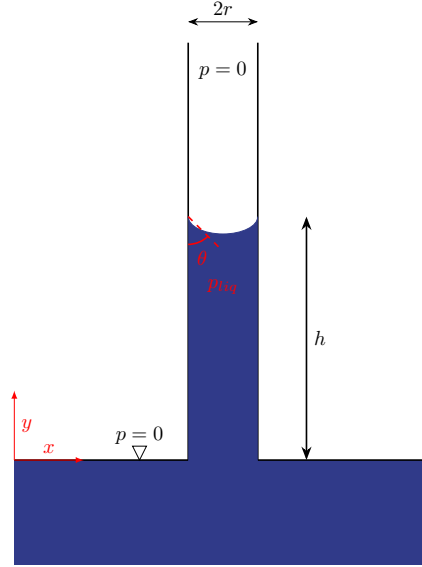


Figure 3: Schematic setup of the cylindrical capillary rise validation benchmark. h denotes the rise height, θ the contact angle, p the gauge pressure and p_{liq} the wetting fluid's pressure

The primary force pulling the fluid up the tube is surface tension acting along the wetted perimeter where the liquid meniscus touches the solid wall. From the Young-Laplace equation (2.1), the capillary pressure jump across a curved spherical meniscus in a 2D tube of radius r is:

$$\Delta p = \frac{\sigma \cos(\theta)}{r} \quad (3.1)$$

To evaluate the robustness of the finite volume solver across different surface wetting conditions and geometric scales, the validation case was executed across a range of static contact angles, $\theta = \{20^\circ, 40^\circ, 60^\circ\}$, as well as two separate tube radii, $r = \{1 \text{ mm}, 0.5 \text{ mm}\}$. According to Equation 3.1, increasing the contact angle reduces the driving force, which should result in a lower equilibrium height. This static equilibrium height, h , is established when the upward capillary pressure is perfectly balanced by the downward hydrostatic pressure of the fluid column ($\Delta p = \rho g h$). By equating these forces, the theoretical rise height can be expressed as:

$$h = \frac{\sigma \cos(\theta)}{\rho g r} \quad (3.2)$$

With the validation framework established, the two-phase simulations were executed and allowed to run until the fluid interface reached static equilibrium. The physical properties assigned to the fluids mirror a standard water-air system, with a surface tension $\sigma = 0.072 \text{ N/m}$, a wetting phase density of $\rho_{\text{water}} = 1000 \text{ kg/m}^3$, a non-wetting phase density of $\rho_{\text{air}} = 1.225 \text{ kg/m}^3$, and kinematic viscosities of $\nu_{\text{water}} = 1 \times 10^{-6} \text{ m}^2/\text{s}$ and $\nu_{\text{air}} = 1.48 \times 10^{-5} \text{ m}^2/\text{s}$. Figure 4 presents the temporal evolution of the interface height for the case $\theta = 20^\circ$ and $r = 1 \text{ mm}$:

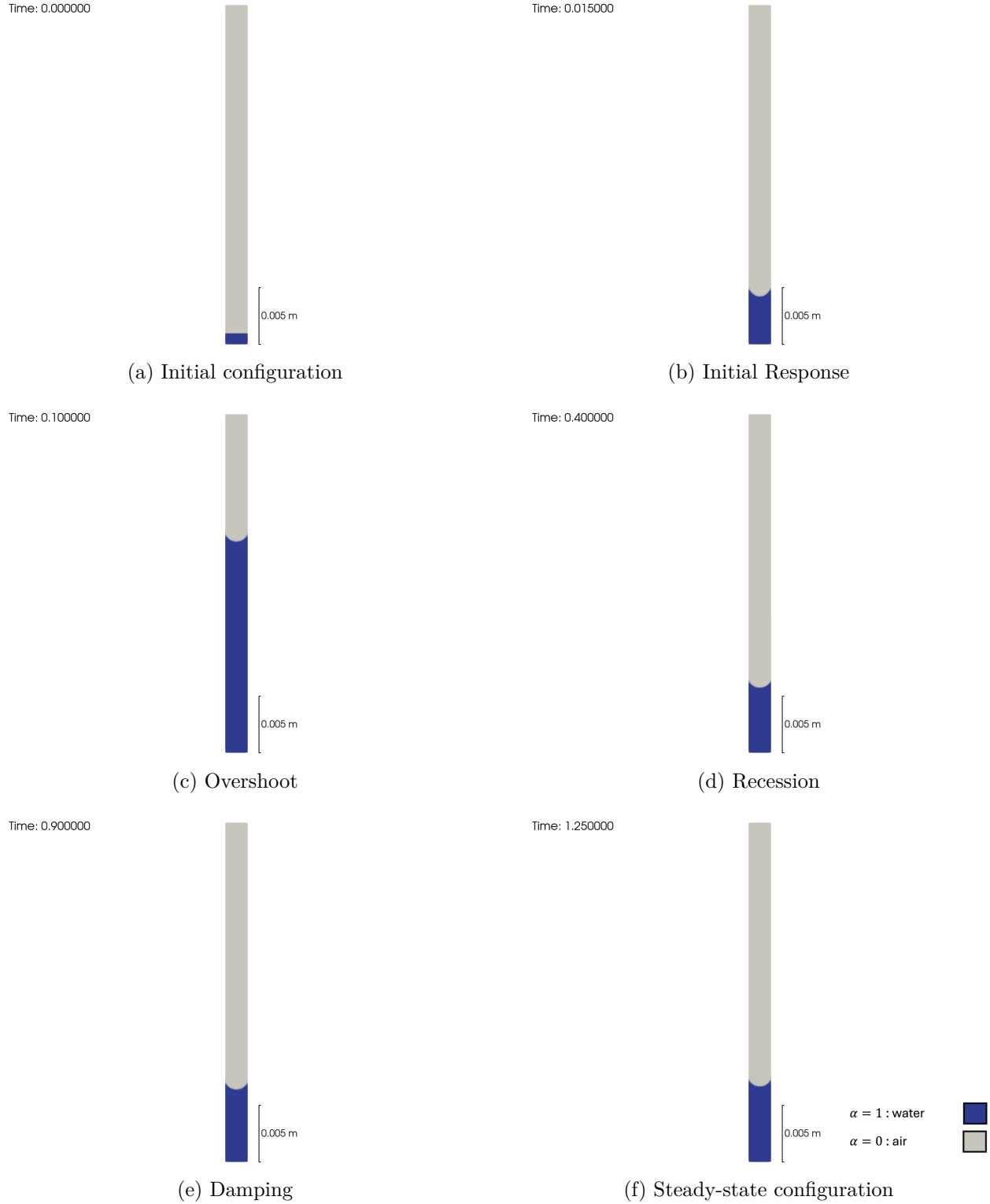


Figure 4: Temporal evolution of a capillary rise interface profile ($\theta = 20^\circ$, $r = 1$ mm)

As shown in Figure 4, the numerical solver successfully captures the complete transient dynamics of the capillary rise. Under the initial conditions at $t = 0$ s, the fluid interface remains perfectly flat, having not yet initiated its upward advancement along the solid boundary (Figure 7a). However,

within a hundredth of a second, the capillary forces established by the $\theta = 20^\circ$ wetting condition drive the liquid upward along the tube wall, initiating the column's ascent (Figure 7b). Within a tenth of a second, the fluid accumulates significant kinetic energy; its upward inertia carries the interface past the true equilibrium mark, culminating in a distinct dynamic overshoot peak (Figure 4c). Past this initial inertial overshoot, the system undergoes gradual recession and viscous damping before achieving a fully stabilized steady-state configuration at the analytical equilibrium height of $h = 6.8$ mm within 1.25 s, as illustrated in Figure 4f.

The underlying physics remain identical across the remaining cases; however, configurations with larger contact angles exhibit a smaller overshoot due to the higher relative influence of viscous damping and lower initial momentum, whereas narrower tube radii yield a more pronounced initial overshoot owing to the intensified capillary pressure driving force. The comprehensive transient evolution plots and equilibrium profiles for these remaining variations are provided in Appendix 6.3 for completeness.

To evaluate the validity of the numerical simulations, the pressure discontinuity across the meniscus at the center of the tube, along with the final equilibrium rise height, were measured at steady state. Tabulated below are the analytical capillary pressures and rise heights compared against the numerical values extracted from the simulations, alongside the corresponding relative errors computed for each configuration:

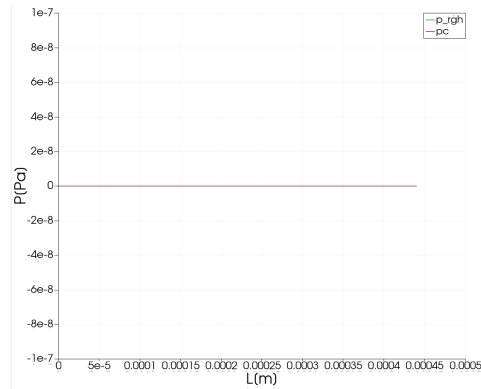
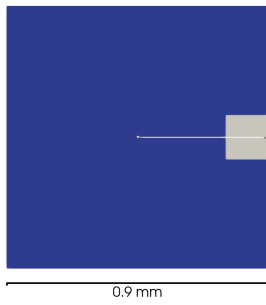
Contact Angle [°]	Tube radius r [mm]	Capillary pressure P_c (analytical) [Pa]	Capillary pressure P_c (simulation) [Pa]	Rise height h (analytical) [mm]	Rise height h (simulation) [mm]	Error [%]
20	1	67.657	64.872	6.68	6.52	4.11
40	1	55.155	52.526	5.62	5.35	4.76
60	1	36.000	34.021	3.67	3.47	5.49
20	0.5	135.310	129.183	13.8	13.17	4.52

Table 1: Comparison of analytical and simulated capillary pressure and equilibrium rise height for different contact angles.

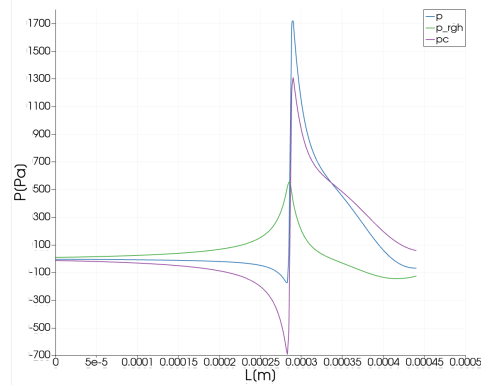
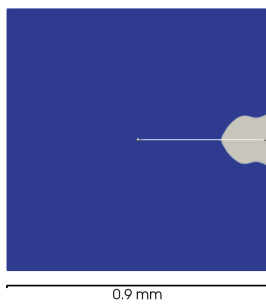
By comparing the analytical and numerical results presented in Table 1, it can be observed that the relative error remains close to 5% across all simulation cases. Considering that these results were obtained using a relatively coarse computational grid, the level of precision is good. Overall, the results provide strong validation of the solver's capability to accurately capture capillary-driven pressure fields and the transient dynamics of fluid interfaces.

3.1.2 Wettability Effects on Trapped Bubble

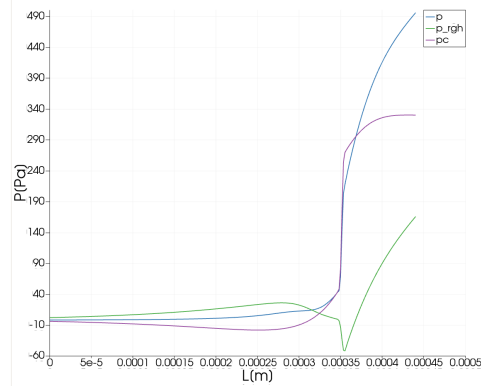
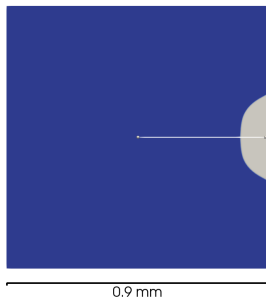
Time: 0.000000s



Time: 0.000100s



Time: 0.000500s



Time: 0.003000s

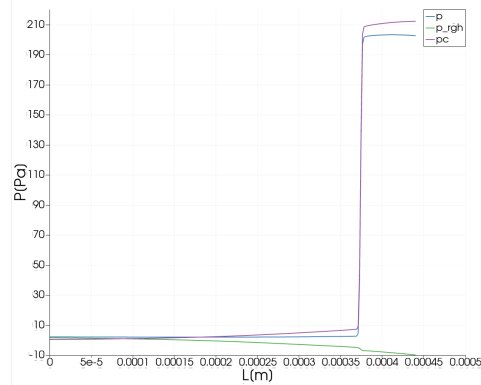
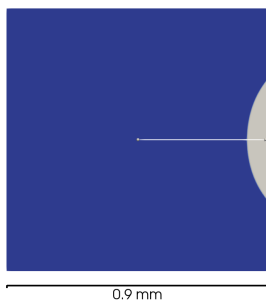
 $\alpha = 1$: water $\alpha = 0$: air

Figure 5: Temporal evolution of a trapped bubble against a solid wall ($\theta = 140^\circ$), plotted alongside the corresponding capillary pressure jump at the interface.

As depicted in Figure 5 and in Section 6.4 where we have all whole steady state configuration of all the other contact angles presented in Table 2, the temporal evolution of the air bubble reveals the transition of the system from its initial state to equilibrium. In this case, the contact angle is set to 140° . Because the solid surface is hydrophobic, the non-wetting fluid (air) is favored, allowing it to occupy a larger contact area along the wall compared to the water. During the early transient phase, there is a pressure jump at the phase boundary, which forces the rapid displacement of the fluids. To minimize the surface tension energy, the interface deforms to adopt a spherical shape. As the simulation progresses the system reaches a steady state ($t = 0.0030$) where the interface geometry stabilizes into a spherical cap. To quantify this behavior, the pressure gradient is extracted along a vector visible in the middle of each frame, which crosses the gas-liquid interface at a single, distinct point. At the final stage, the capillary pressure p_c successfully equilibrates, which is demonstrated by the constant pressure jump across the interface in the corresponding plot. We still observe minor pressure oscillations for the hydrodynamic pressure p_{rgh} , which are numerical errors known as spurious currents. They originate from an error in calculating the curvature, and create artificial velocities.

Contact Angle [°]	Measured Contact Angle [°]	Radius of the Bubble [m]	Capillary pressure P_c (analytical) [Pa]	Capillary pressure P_c (simulation) [Pa]	Error [%]
0	0	84.8×10^{-6}	849.1	840	1.07
60	58.3	94.5×10^{-6}	761.6	705	7.43
90	90	119.9×10^{-6}	600.4	570	5.06
140	134.5	331.3×10^{-6}	217.3	230	5.52
180	$\emptyset$	∞	0	0	$\emptyset$
90*	90	11.99×10^{-6}	6005	5999	0.1

Table 2: Comparison of analytical and simulated capillary pressure for different contact angles.

*This domain is smaller by a factor of 10.

To validate the numerical simulations, we compared the simulated capillary pressure with the analytical results for a trapped air bubble attached to the wall.

For the analytical derivation, we assume that the air bubble is completely incompressible, meaning its cross-sectional area S (in our 2D representation) remains constant. Based on the geometric relationship between the bubble's area, its radius R , and the contact angle θ , we have:

$$S = R^2(\pi - \theta + \sin(\theta) \cos(\theta)) \quad (3.3)$$

By rearranging this equation, we can express the radius of the bubble R as a function of its constant area and the contact angle:

$$R = \sqrt{\frac{S}{\pi - \theta + \sin(\theta) \cos(\theta)}} \quad (3.4)$$

Finally, we apply the Young-Laplace equation. By substituting the geometric radius R into the capillary pressure formula, we obtain the analytical expression for the capillary pressure P_c :

$$P_c = \sigma \sqrt{\frac{\pi - \theta + \sin(\theta) \cos(\theta)}{S}} \quad (3.5)$$

where σ represents the surface tension between the two fluids. Therefore, we can see in Table 2 that the error remains below a range of 7.5% when the scale of the experiment is in the order of 1mm. However, the error margin becomes completely negligible when we dive into smaller scales. These observations coincide with the microscale expectations of GeochemFoam.

3.2 Complex Porous Media Geometries

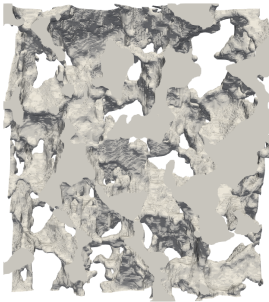
Having successfully validated the numerical solver's accuracy against analytical solutions, we can move on to more complex geometries in order to simulate fluid infiltration in complex porous media.

3.2.1 Bentheimer Sandstone

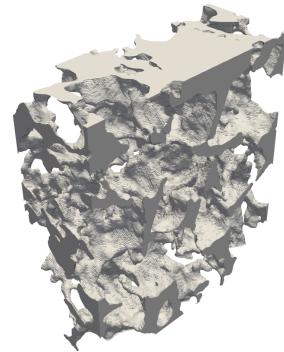
Initially, the goal was to investigate moisture transport in cement-based materials. However, such materials exhibit a highly disordered, multi-scale pore framework spanning from nanometer-scale gel pores to millimeter-scale capillary voids. Capturing continuous fluid-fluid interfaces across these vast multi-scale boundaries demands extremely fine spatial discretization, making simulations extremely time-consuming and computationally expensive.

Consequently, to maintain numerical feasibility within the time constraints of this project, the primary focus was redirected to Bentheimer Sandstone. Bentheimer Sandstone provides an ideal replacement for cement-based materials; it features a significantly more uniform and well-characterized micrometer-scale resolution ($5 \mu\text{m}$) compared to the disordered, multi-scale network of cement paste, while preserving the critical geometric complexities inherent to natural porous media. The 3D microstructural domain used for the multi-phase transport simulations in this project was constructed using the benchmark **Bentheimer400- $5 \mu\text{m}$** X-ray Micro-Computed Tomography (micro-CT) dataset. This raw binary volume data, representing a 400^3 voxel cube at a spatial resolution of $5 \mu\text{m}$, was sourced from the open-access repositories provided by GeoChemFoam [Maes and Menke, 2026].

From the primary $400 \times 400 \times 400$ voxel **Bentheimer400- $5 \mu\text{m}$** domain, two different subdomains were cut out and used for their respective objectives in order to further optimize computational efficiency while maintaining a representative pore-network structure. These subdomains can be seen in Figure 6 and Figure 7:



(a) Cross-sectional view looking down the $+X$ axis



(b) Isometric 3D View

Figure 6: Extracted $50 \times 140 \times 125$ voxel subdomain used for analyzing microscopic wetting dynamics and contact angle variations along the 140×125 walls.

(a) Cross-sectional view looking down the $-Z$ axis

(b) Isometric 3D View

Figure 7: Extracted $80 \times 140 \times 80$ voxel subdomain used for investigating the impact of liquid property variations between two distinct fluids.

3.2.2 Effects of Surface Wettability

The main objective in this part of the project is to study the impact of wettability on a steel surface in the porous scale structures. As we saw previously in the section 2.1.1, the wettability of a surface impact directly the covering area of a solid surface by a fluid. Therefore we placed two steel wall conditions on each side of Bentheimer sandstone (Y-Z plan) to cover a large area to extract the saturation water on each of these surfaces. To properly evaluate these effects, we run two independent simulations. In the first simulation, the contact angle of the steel walls is set to 60° to represent a hydrophilic surface. In the second simulation, the walls are set to a hydrophobic contact angle of 120° . In both simulations, the respective contact angles of the sandstone remain the same throughout the geometry (20°).

By visually comparing the temporal evolution of the fluid in Figure 8 and 18 (hydrophilic case) and Figure 9 and 19 (hydrophobic case), distinct infiltration patterns immediately emerge. In the $\theta = 60^\circ$ scenario, the water demonstrates a strong affinity for the steel boundary. Driven by capillary forces, the fluid rapidly spreads upwards and creeps along the solid surface, filling the adjacent pore spaces extensively. Inversely, in the $\theta = 120^\circ$ scenario, the non-wetting nature of the wall visibly repels the water. The fluid interface bulges away from the steel boundary to minimize contact area, significantly restricting the upward capillary rise and leaving large sections of the wall exposed to the secondary phase (air).

Time: 0.000000



Initial configuration

Time: 0.000500



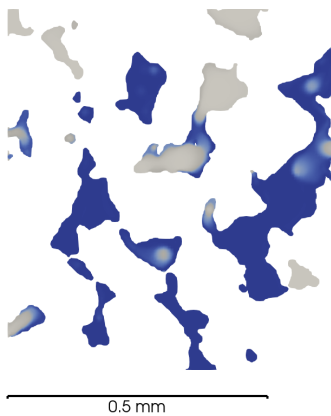
Time: 0.001500



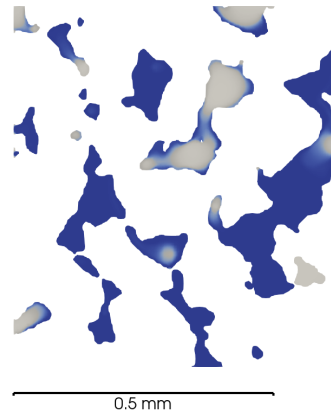
Time: 0.002500



Time: 0.005000



Time: 0.015000



Steady-state configuration

Figure 8: Temporal evolution of water rise along the front steel wall for a contact angle ($\theta = 60^\circ$)

Time: 0.000000



Time: 0.000500



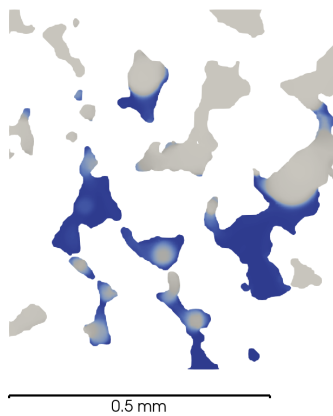
Time: 0.001500



Time: 0.002500



Time: 0.005000



Time: 0.015000



Figure 9: Temporal evolution of water rise along the front steel wall for a contact angle ($\theta = 120^\circ$)

To quantify these qualitative visual observations, we calculated the water saturation (S_w) specifically at these boundary walls. As defined previously, saturation represents the percentage of the total available pore area on that specific plane that is effectively filled with water, quantifying the local fluid distribution:

$$S_w = \frac{\text{Area}_w}{\text{Area}_{\text{tot}}} = \frac{\text{Area}_w}{\text{Area}_w + \text{Area}_a}$$

where Area_w and Area_a are the areas occupied by water and air, respectively. By isolating the two steel walls, we can plot the temporal evolution of this saturation to directly compare the impact of surface wettability.

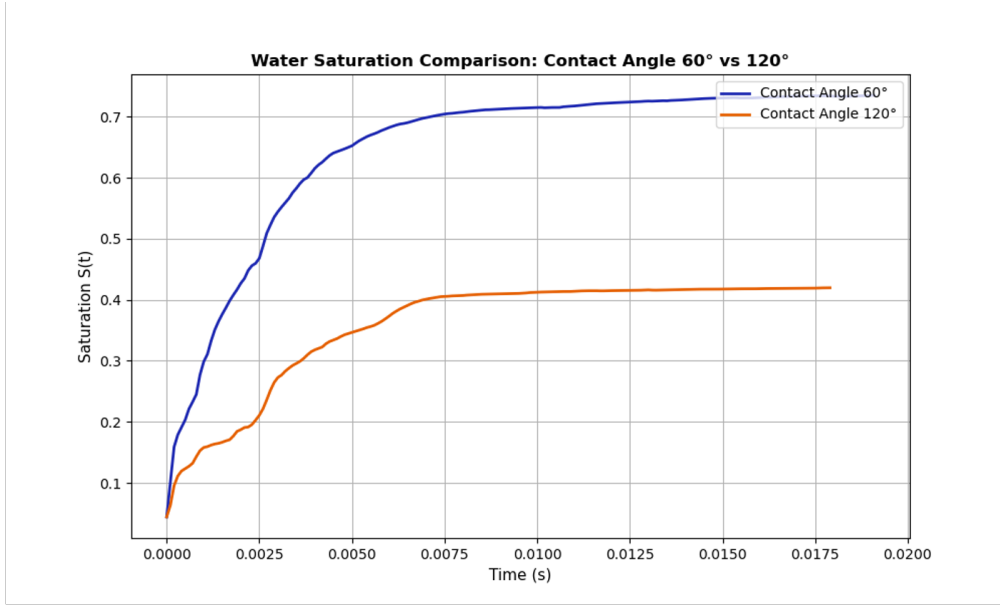


Figure 10: Temporal evolution of water saturation (S_w) at the steel boundary walls for hydrophilic ($\theta = 60^\circ$) and hydrophobic ($\theta = 120^\circ$) contact angles.

As shown in Figure 10, the quantitative results align perfectly with our theoretical expectations and previous visual observations. During the initial phase of the simulation ($t < 0.0050$ s), both configurations exhibit a rapid increase in water saturation as the fluid begins to infiltrate the bottom of the structure. The minor discontinuities visible during this steep rise are due to the 2D nature of the surface measurement. Because the solid rock matrix separates individual pores along the wall, the saturation increases in discrete steps as the fluid sequentially fills these distinct pore spaces. However, the curves quickly diverge based on the enforced surface wettability.

For the hydrophilic case ($\theta = 60^\circ$), the strong capillary forces promote continuous fluid imbibition along the steel wall. The water is actively drawn upward, allowing the saturation to reach a high steady-state value of approximately 75%. On the other hand, the hydrophobic case reaches a saturation of 42% because of the non-wetting nature of the boundary resulting in an opposing capillary force. This repulsion physically limits the extent of penetration of the fluid because the water cannot infiltrate the region close to the wall.

Nevertheless, despite the difference in wettability along the steel wall, the saturation plateau is reached at the same time (≈ 0.0075 s).

3.2.3 Water VS IPA

As stated previously, the second goal of this project was how the distinct properties of water and isopropyl alcohol impact fundamental fluid behaviors within porous media. To achieve this, two independent simulations were performed using distinct fluids as benchmarks: water and isopropyl alcohol (IPA). The physical properties of both these phases are detailed below:

	Water	Isopropyl Alcohol (IPA)
Surface Tension Liquid-Air [N/m]	0.072	0.022
Density [kg/m³]	1000	785
Kinematic viscosity [m²/s]	1×10^{-6}	2.2×10^{-6}
Contact Angle [°]	20	5

Table 3: Physical properties of the benchmark fluid phases used during the simulations

The physical parameters outlined in Table 3 establish a clear difference in how the two fluids will move through the pore spaces. Most notably, while water exhibits a high surface tension ($\sigma = 0.072$ N/m) and a lower kinematic viscosity, IPA presents nearly a threefold reduction in surface tension ($\sigma = 0.022$ N/m) alongside more than double the kinematic viscosity ($\nu = 2.2 \times 10^{-6}$ m²/s). Furthermore, the prescribed contact angles ($\theta = 20^\circ$ for water and $\theta = 5^\circ$ for IPA) imply different wetting states along the silica pore network. Consequently, evaluating these two distinct fluids allows for a systematic investigation into how competing capillary pressures and viscous forces influence the overall dynamics within the irregular pore network.

At the start of the simulation ($t = 0$ s), identical initial and boundary conditions were enforced across both fluid test cases to ensure a controlled numerical comparison. The irregular Bentheimer sandstone subdomain is initially primed primarily with the defending gas phase ($\alpha = 0$). However, to prevent nonphysical numerical instabilities and simplify the phase fraction gradient calculations at the bottom boundary patch, a thin buffer layer of approximately 0.0001 m of the respective invading liquid ($\alpha = 1$) is pre-allocated along the Y-axis inlet zone, as shown in Figure 11:

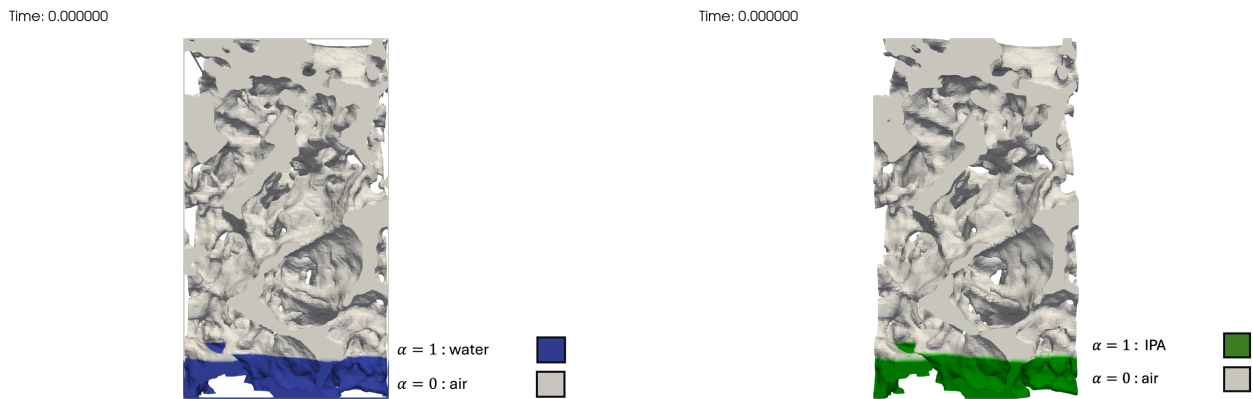
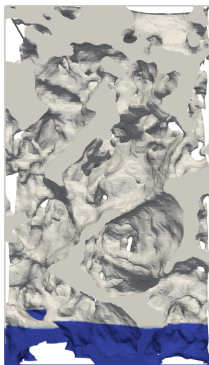


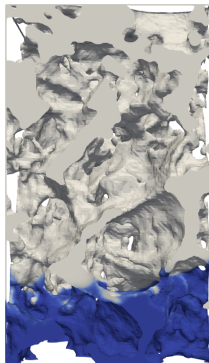
Figure 11: Initial conditions for the simulations

The following figures (Figures 12 and 13) represent the temporal evolution of both liquids: From the visual results, we already see that the water phase rises much faster within the pores than

Time: 0.000000

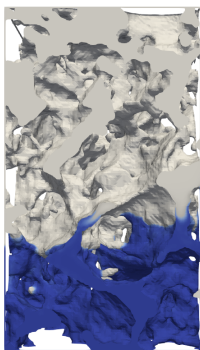


Time: 0.000100

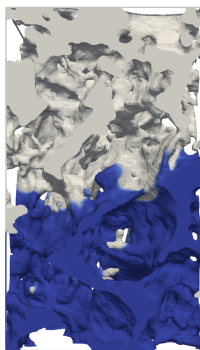


Initial configuration

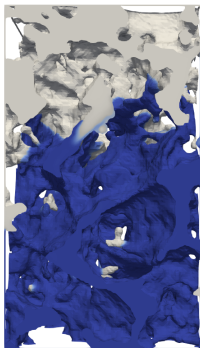
Time: 0.000400



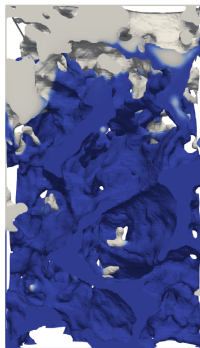
Time: 0.001200



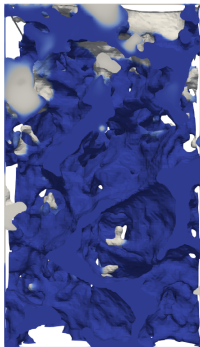
Time: 0.002400



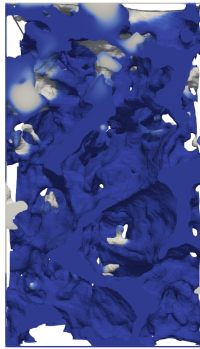
Time: 0.004000



Time: 0.006000



Time: 0.015000



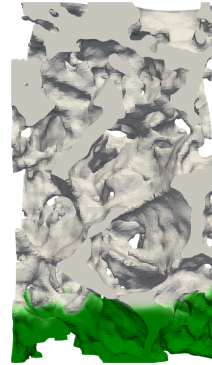
Steady-state configuration

Figure 12: Temporal evolution of Water within Bentheimer Sandstone

Time: 0.000000

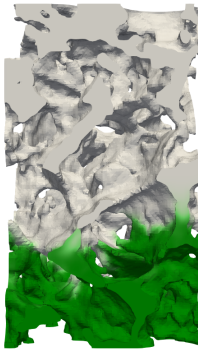


Time: 0.000100

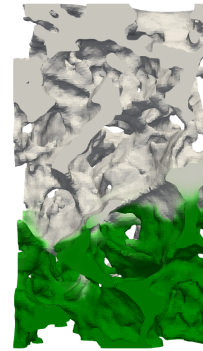


(a) Initial configuration

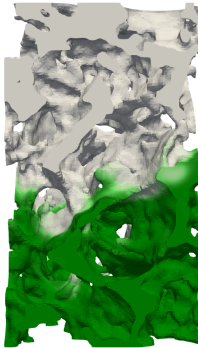
Time: 0.001200



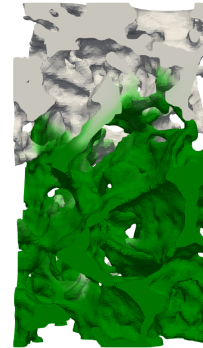
Time: 0.002400



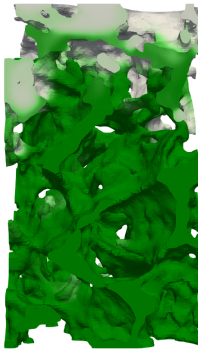
Time: 0.004000



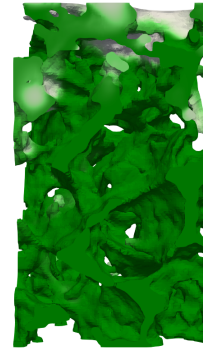
Time: 0.010000



Time: 0.020000



Time: 0.034000



Steady-state configuration

Figure 13: Temporal evolution of IPA within Bentheimer Sandstone

the IPA. However, the steady-state configuration figures do not provide a definitive measurement of how much fluid has actually penetrates the internal pore network. To quantify these results, we introduce a new quantity called the liquid saturation S .

In the context of multiphase flow through porous media, the liquid saturation represents the volumetric fraction of the open pore space that is occupied by the invading fluid phase. It is mathematically defined as:

$$S_{\text{liquid}} = \frac{V_{\text{liquid}}}{V_{\text{pore}}} \quad (3.6)$$

The temporal evolution of the overall phase saturation for both the water and IPA cases is plotted in Figure 14:

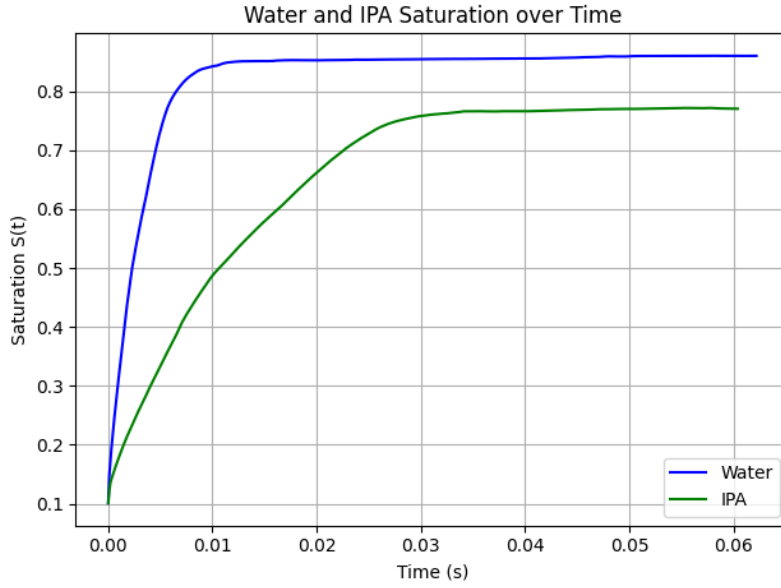


Figure 14: Comparison of the transient liquid saturation S_{liquid} as a function of simulation time for the water (blue) and IPA (green)

The quantitative results presented in Figure 14 clearly demonstrate a significant difference in the saturation dynamics between the two fluid cases. For the water phase (blue curve), a rapid, quasi-linear initial invasion rate is observed. The initial saturation gradient drives the liquid volume fraction to approximately 85% within the first 0.01 s of the simulation. Following this rapid transient surge, the water saturation profile undergoes a sharp inflection, abruptly stabilizing into a flat steady-state plateau at a maximum value of roughly 86%.

Conversely, the Isopropyl Alcohol (IPA) phase (green curve) exhibits a more gradual accumulation rate. In this case, the saturation increases smoothly, ultimately leveling off at a lower steady-state maximum of approximately 77%. Furthermore, the IPA front requires nearly three times as long as the water phase to reach this stabilization milestone, taking approximately 0.035 s to fully establish its plateau.

4 Discussion

4.1 Interpretation

4.1.1 Impact of Surface Wettability

Modifying the wettability of the boundary walls dictates the local flow dynamics, impacting the overall fluid distribution and infiltration capacity within the adjacent porous structure. The boundary wettability simulations separate flow kinetics and final fluid distribution. Both the hydrophilic (60°) and hydrophobic (120°) cases reach steady state at the same time ($t \approx 0.0075$ s). This shows that the stabilization time is governed by the global fluid properties and domain size, rather than local wall wettability.

However, the contact angle controls the fluid covering along the surface. A hydrophilic surface acts as a preferential pathway. Capillary forces pull water along the wall, overcoming local pore resistance to achieve 75% saturation. In contrast, a hydrophobic surface repels the fluid. This minimizes solid contact, traps air in adjacent pores, and restricts total saturation to 42%.

Since solid boundaries in sandstone act as hydrophilic pathways for moisture, applying targeted hydrophobic treatments can block local capillary water ingress while leaving the surrounding matrix unmodified.

4.1.2 Role of Fluid Properties (Water vs. IPA)

The distinct saturation dynamics observed between the two liquids highlight the profound impact of intrinsic fluid properties on invasion mechanisms within the porous medium. The rapid, quasi-linear initial invasion of the water phase (85% within 0.01 s) indicates a regime where the saturation evolution is primarily driven by surface tension-dominated capillary forces at these small scales. Because water possesses a significantly higher surface tension than Isopropyl Alcohol (IPA), it generates a larger capillary pressure difference (P_c) at the fluid-fluid interface, aggressively pulling the liquid into the pore network.

Crucially, the differences in transport kinetics are further exacerbated by viscosity. The lower kinematic viscosity of water leads to reduced flow resistance within the tortuous pathways, facilitating a faster capillary-driven invasion.

On the other hand, the IPA phase exhibits a more gradual accumulation and a lower steady-state plateau (77%). This behavior is governed by the compounding effects of its lower surface tension, which weakens the primary capillary driving force, and its higher kinematic viscosity, which increases viscous drag. This heightened flow resistance directly accounts for the prolonged stabilization time (0.035 s), which is nearly three times longer than that of water. Furthermore, the 9% deficit in the final saturation plateau for IPA indicates that this less favorable balance of forces fundamentally alters the pore-scale invasion behavior. Because the reduced capillary pressure is insufficient to overcome the higher entry pressures of the most constricted pore throats, the IPA phase cannot penetrate the finest capillary networks. Because of the weaker driving force and higher flow resistance, the invading fluid struggles to displace the original fluid evenly. Instead, it frequently breaks apart or flows around it. Consequently, large pockets of the original fluid become permanently trapped in the pore network, limiting the final saturation level.

4.2 Assumption and Source of Errors

To evaluate the reliability of the final simulations within the Bentheimer sandstone, the fundamental assumptions and sources of error must be outlined.

Physical Assumptions:

- Incompressible fluids: The invading liquids (water, IPA) and the air are treated as incompressible. Densities and kinematic viscosities remain constant.
- Isothermal conditions: Temperature variations are neglected. Properties such as surface tension do not change during the simulation.
- Rigid rock matrix: The sandstone micro-geometry is assumed perfectly rigid and does not deform under fluid pressure.
- Static contact angles: Constant contact angles are prescribed at the boundaries.

Sources of Error:

- Lack of mesh refinement study: Due to time constraints, parallel simulations with a refined mesh could not be executed. A formal mesh independence study was not performed to verify the reliability of the grid resolution on the final saturation plateaus.
- Voxel resolution limits: The $5\text{ }\mu\text{m}$ resolution of the micro-CT dataset cannot resolve sub-grid surface roughness or pore throats smaller than the voxel size. This limits the accuracy of the local capillary entry pressure.
- Spurious currents: The numerical solver generates unphysical velocity vectors (spurious currents) near the fluid-fluid interface due to minor errors in calculating the local surface curvature.
- Interface smearing: The volume fraction method smears the phase boundary over a few computational cells. This reduction in interface sharpness creates minor uncertainties in the internal volume and saturation calculations.

5 Conclusions

Throughout this semester project, we learned to use OpenFOAM and GeoChemFoam for multiphase flow simulations. The initial reference cases allowed us to understand the implementation of key parameters, including capillary pressure, surface tension, contact angles, and boundary conditions. These validation tests established the foundation required to operate the software.

We applied these skills to simulate fluid infiltration within a complex Bentheimer sandstone microstructure. Two distinct investigations were performed. On one side, we compared the infiltration dynamics of water and Isopropyl Alcohol (IPA) to evaluate the impact of fluid properties such as viscosity and surface tension by extracting the volume saturation. On the other side, we imposed hydrophilic (60°) and hydrophobic (120°) contact angles on steel boundary walls to extract surface water saturation data and observe local fluid distribution.

For future improvements, these simulations could be executed across different mesh resolutions to conduct a formal grid independence study and verify the precision of the final saturation data. Furthermore, future work could extend this framework to reactive multiphase transport. This would enable the study of chemical reactions, corrosion, and material degradation as a function of local fluid saturation and surface wettability. But since material degradation physically alters the solid matrix, such advanced models would need to account for dynamic changes in pore geometry and surface wettability over time.

Acknowledgments

We would like to thank Dr. Luka Malenica for his continuous guidance and weekly feedback throughout this project. His support allowed us to significantly develop our skills in using CFD software for multiphase flow in porous media. We also thank Prof. Ueli Angst for his supervision and for welcoming us as mechanical engineering students within the D-BAUG department. Finally, we thank the entire research team and the department for their welcome and for providing all the necessary equipment (especially the servers and the meeting room HIF E 28).

References

- [1] Booyabazooka. Water droplets and fluid phase boundary illustration, November 2008. URL <https://commons.wikimedia.org/wiki/File:Wassermolek%C3%BCleInTr%C3%B6pfchen.svg>.
- [2] Art Gatenby. Initiation to Contact Angle, 2016. URL <https://www.cscscientific.com/csc-scientific-blog/initiation-to-contact-angle>.
- [3] Dhanalakshmi K, Maheswaran. J, Siva Avudaiappan, Mugahed Amran, Radhamanohar Aepuru, Roman Fediuk, and Nikolai Vatin. Prediction of Pore Volume Dispersion and Microstructural Characteristics of Concrete Using Image Processing Technique. *Crystals*, 11(12):1476, November 2021. ISSN 2073-4352. doi: 10.3390/cryst11121476. URL <https://www.mdpi.com/2073-4352/11/12/1476>.
- [4] Julien Maes and Hannah Menke. GeoChemFoam: Multi-phase reactive transport solver, version 5.2, June 2026. URL <https://github.com/GeoChemFoam/GeoChemFoam>. original-date: 2021-10-18T11:12:31Z.
- [5] F. Moukalled, L. Mangani, and M. Darwish. *The finite volume method in computational fluid dynamics: an advanced introduction with OpenFOAM and Matlab*. Number 113 in Fluid Mechanics and Its Applications. Springer, Cham Heidelberg New York Dordrecht London, 2016. ISBN 9783319168739 9783319168746.
- [6] OpenCFD Ltd. OpenFOAM v2212 User Guide, 2022. URL <https://www.openfoam.com/documentation/user-guide>.
- [7] Edward W. Washburn. The Dynamics of Capillary Flow. *Physical Review*, 17(3):273–283, March 1921. ISSN 0031-899X. doi: 10.1103/PhysRev.17.273. URL <https://link.aps.org/doi/10.1103/PhysRev.17.273>.
- [8] Zhidong Zhang and Ueli Angst. Modeling Anomalous Moisture Transport in Cement-Based Materials with Kinetic Permeability. *International Journal of Molecular Sciences*, 21(3):837, January 2020. ISSN 1422-0067. doi: 10.3390/ijms21030837. URL <https://www.mdpi.com/1422-0067/21/3/837>.
- [9] B.V. Zhmud, F. Tiberg, and K. Hallstensson. Dynamics of Capillary Rise. *Journal of Colloid and Interface Science*, 228(2):263–269, August 2000. ISSN 00219797. doi: 10.1006/jcis.2000.6951. URL <https://linkinghub.elsevier.com/retrieve/pii/S0021979700969510>.

6 Appendix

6.1 Project Contributions

The authorship of the project is as follows:

Villa César Jacques Hanakea Fomalhaut Melchior:

- Abstract
- [1.1](#) Motivation (Literature review)
- [2.1](#) Flow Physics and Definitions
 - [2.1.1](#) Surface Tension, Contact angle and Wettability (contact angle and Wettability part)
 - [2.1.3](#) Governing Equations
- [2.2.1](#) VOF
- [2.2.3](#) Courant Number
- [3.1.2](#) Wettability Effects on Trapped Bubble
- [3.2.2](#) Effects of Surface Wettability
- [4.1.1](#) Impact of Surface Wettability
- [4.2](#) Assumption and Source of Errors
- [5](#) Conclusion

Baltaian Nicholas:

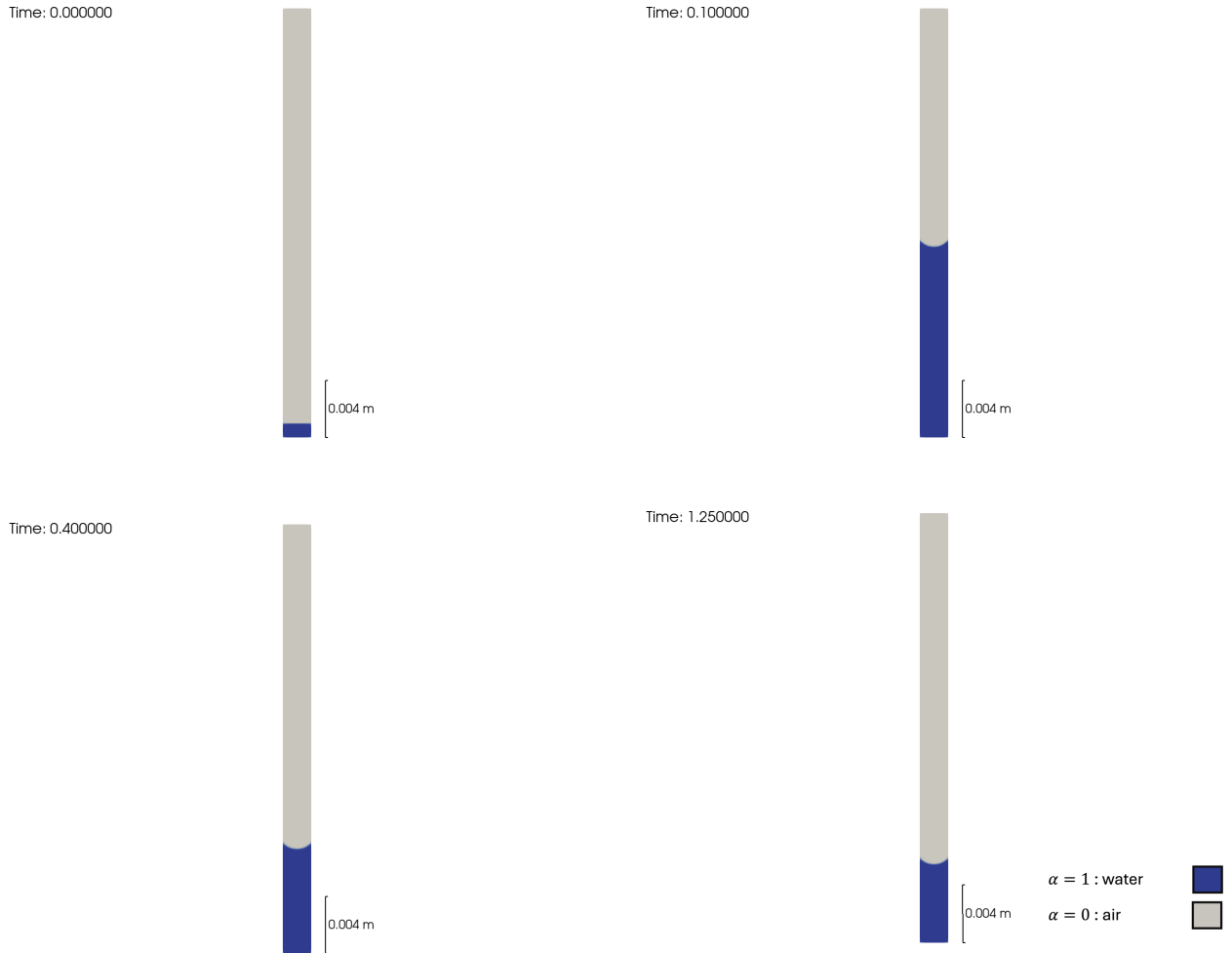
- [1.1](#) Motivation
- [1.2](#) Project Objectives
 - [2.1.1](#) Surface Tension, Contact angle and wettability (surface tension part)
 - [2.1.2](#) Young-Laplace equation and capillary pressure
- [2.2.2](#) Finite Volume Method
- [3.1](#) Validation Cases on Simple Geometries
 - [3.1.1](#) Capillary Rise in a Tube
- [3.2](#) Complex Porous Media Geometries
 - [3.2.1](#) Bentheimer Sandstone
 - [3.2.3](#) Water VS IPA
- [4.1.2](#) Role of Fluid Properties (Water vs. IPA)

6.2 Supplementary Simulation Interface Frames

6.3 Capillary Rise

The visual frame sequences captured from the remaining configurations within the validation parameters are presented below. While the final steady-state capillary pressures for these cases are comprehensively analyzed and tabulated in the main body of the text, these visual trajectories are included for completeness.

As illustrated in the following frame matrices, the transient interface progression follows the same fundamental phases of initial wall wetting and upward advancement seen in the primary case. However, a clear physical variation is visible: as the contact angle increases ($\theta = 40^\circ, 60^\circ$), the characteristic dynamic overshoot is significantly suppressed. These sequences visually confirm how the relative intensification of viscous shear forces along the tube boundaries rapidly damps the fluid column's momentum, leading to a smoother, more direct stabilization at the final equilibrium line, as illustrated in the following frame matrices:



(a) Steady-state configuration

Figure 15: Supplementary simulation interface configurations ($\theta = 40^\circ, r = 1\text{mm}$), highlighting the smaller inertial overshoot due to lower surface tension force

Figure 16: Supplementary simulation interface configurations ($\theta = 60^\circ, r = 1\text{mm}$)

On the other hand, when the radius decreases to $r = 0.5$ mm, we see a more pronounced initial overshoot owing to the intensified capillary pressure driving force.

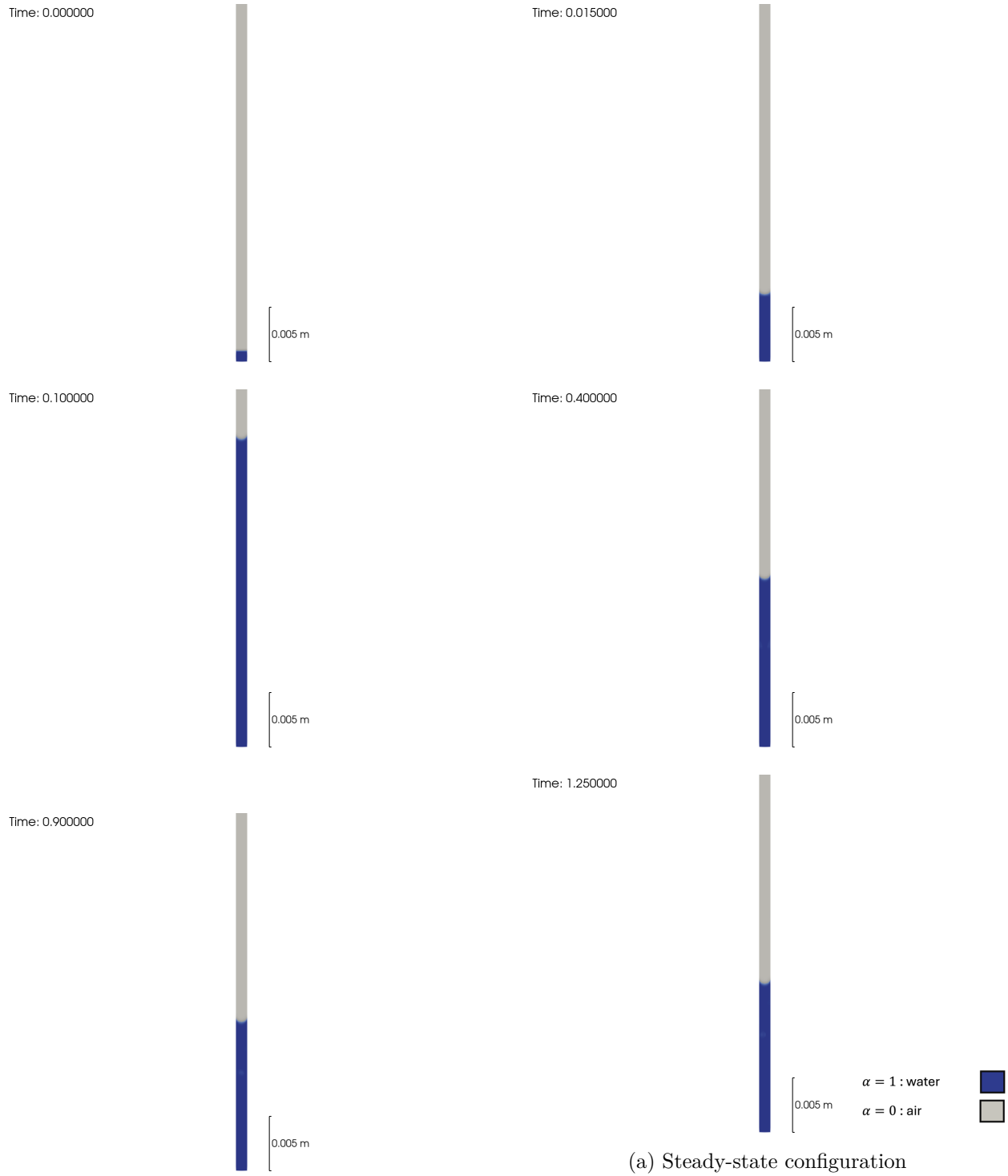
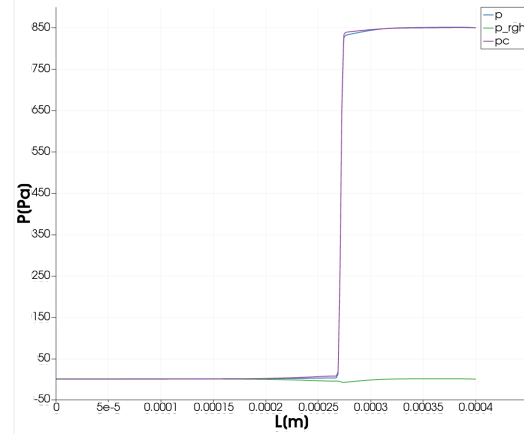
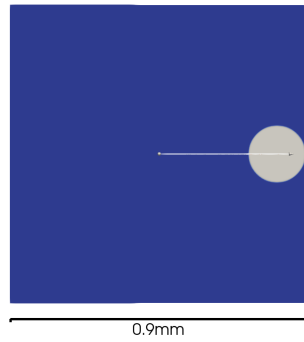


Figure 17: Temporal evolution of a capillary rise interface profile ($\theta = 20^\circ$, $r = 0.5$ mm)

6.4 Effect of Wettability

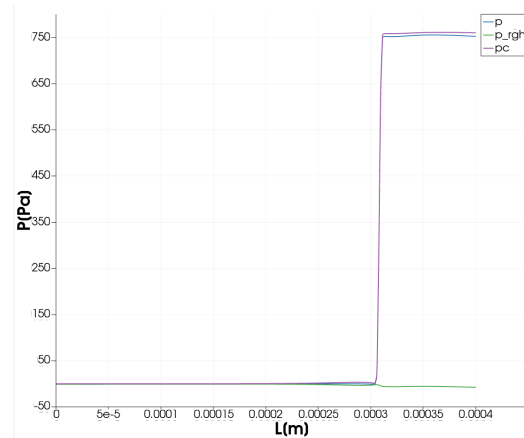
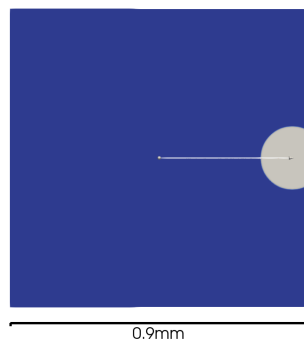
Below we can see the steady state configuration of each bubble with the different contact angles mentioned in Table 2.

Time: 0.020000



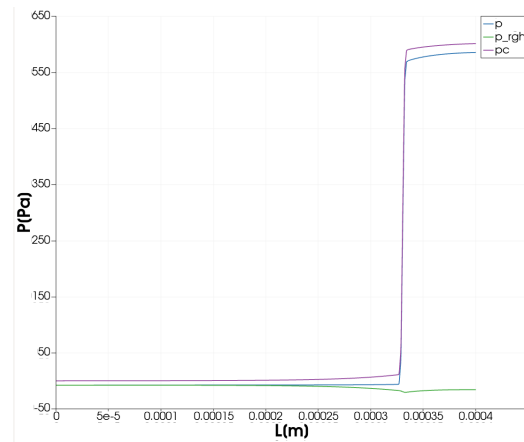
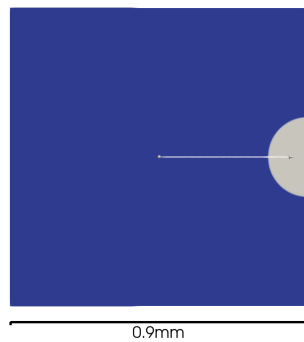
Steady state bubble and pressure profile for a contact angle of $\theta = 0^\circ$.

Time: 0.020000



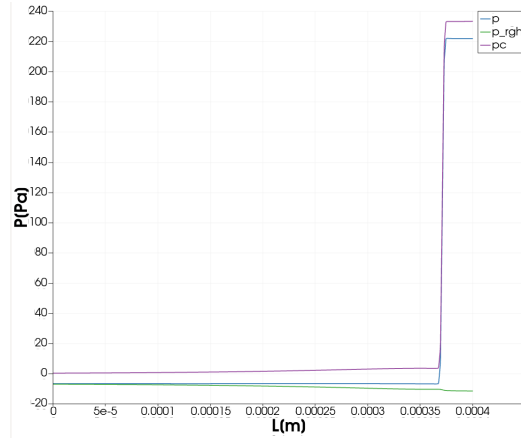
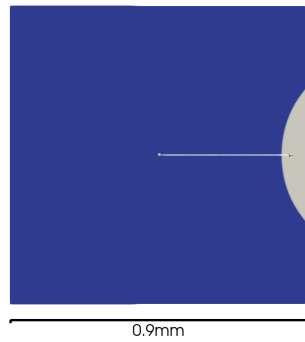
Steady state bubble and pressure profile for a contact angle of $\theta = 60^\circ$.

Time: 0.020000



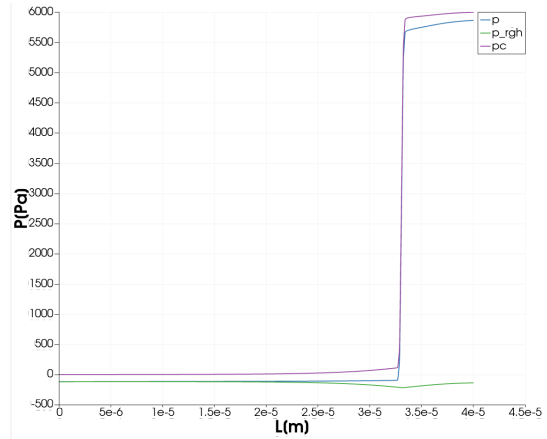
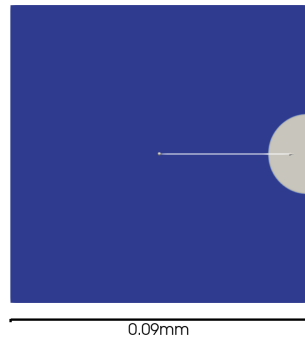
Steady state bubble and pressure profile for a contact angle of $\theta = 90^\circ$.

Time: 0.020000



Steady state bubble and pressure profile for a contact angle of $\theta = 140^\circ$.

Time: 0.020000



Reduced scale of the steady state bubble and pressure profile for a contact angle of $\theta = 90^\circ$.

In the following sequence of frames, we have the evolution of the rise of water along the steel wall behind. We consider the saturation of the back and front wall to do our saturation in the Graph 10.

Time: 0.000000



Initial configuration

Time: 0.000500



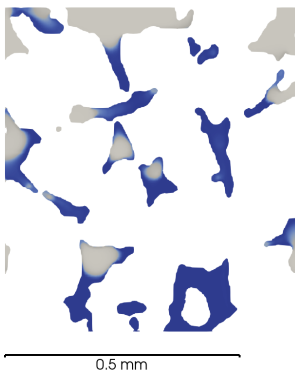
Time: 0.001500



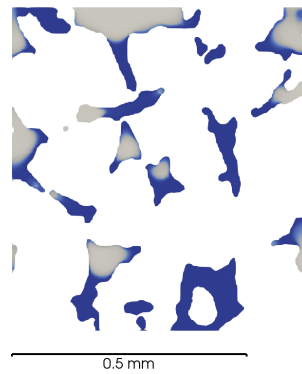
Time: 0.002500



Time: 0.005000



Time: 0.015000



Steady-state configuration

Figure 18: Temporal evolution of water rise along the back steel wall for a contact angle ($\theta = 60^\circ$)

Time: 0.000000



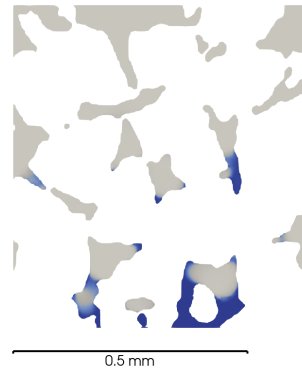
Time: 0.000500



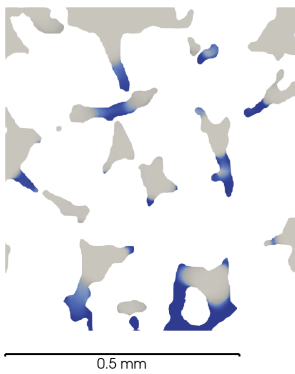
Time: 0.001500



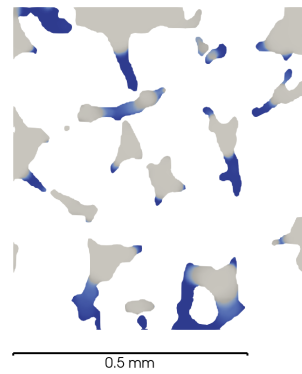
Time: 0.002500



Time: 0.005000



Time: 0.015000

Figure 19: Temporal evolution of water rise along the back steel wall for a contact angle ($\theta = 120^\circ$)